

PROJECT PROGRESS REPORT

GxP Training Bot

AI-Assisted Adaptive Training and Assessment Platform for Pharmaceutical SOPs

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*Report status: progress report covering development from 13 July 2026 to 14 August 2026.
All technical claims and results in this report were verified against the project repository.*

Table of Contents

1. Abstract	4
2. Introduction.....	4
2.1 Limitations of static training	4
2.2 The role of generative AI, and its limits.....	5
2.3 Motivation for SME validation and adaptive learning.....	5
3. Problem Statement.....	5
4. Aim	6
5. Objectives	6
6. Literature Review.....	6
7. Existing System	7
8. Proposed System	8
9. System Architecture.....	8
9.1 Architectural principle: separation of concerns.....	9
10. Development Methodology and Progress.....	10
11. Adaptive Learning Methodology.....	10
11.1 The unit of adaptation.....	10
11.2 Decision pipeline	11
11.3 Online estimation components	12
11.4 Separation of "what" from "when"	12
11.5 Classification of the implemented system.....	13
12. Defects Identified and Corrected During Development	14
Defect 1: Correctly answered sections were still selected for retraining.....	14
Defect 2: Sections never assessed were invisible to selection	14
Defect 3: Whole-document mastery concealed weak sections	14
Defect 4: The answer key was exposed to learners.....	15
Defect 5: A completed assessment attempt could be resubmitted.....	15
Defect 6: Approved content could be modified after signature	15
Defect 7: Duplicate question identifiers fabricated evidence.....	15
Defect 8: Recommendations could not be acted upon	16
13. Testing and Validation	17
13.1 Test distribution by application.....	17
13.2 Categories of testing performed	17
13.3 Controlled adaptive scenarios.....	17
14. Results	18
14.1 Workflow verification	18
14.2 Adaptive selection.....	19

14.3 Measured learning gain.....	19
14.4 System verification results	20
15. Current Project Status.....	21
15.1 Objectively derived completion measure	22
16. Limitations	22
16.1 The offered question set is not persisted	22
16.2 No SOP version lifecycle	23
16.3 Difficulty does not influence adaptive priority	23
16.4 Grounding is provenance, not entailment.....	23
16.5 No formal knowledge-tracing model.....	23
16.6 Further limitations	23
17. Future Scope	24
17.1 Short term	24
17.2 Advanced	24
17.3 Research and agentic direction	25
18. Conclusion.....	25
19. References	26
Appendix A. Progress Summary	27
Appendix B. Progress Timeline	27

1. Abstract

Training on Standard Operating Procedures (SOPs) is a mandatory and recurring obligation in pharmaceutical manufacturing, yet it is typically delivered as static material accompanied by a single generic assessment. Such an approach records that a learner performed poorly but does not identify which part of the procedure was not understood, and consequently retraining repeats content the learner has already demonstrated.

The developed platform, GxP Training Bot, addresses this gap. Controlled SOP documents are ingested, text is extracted, and the document is divided into sections using a three-tier chunking cascade. A large language model, accessed through NVIDIA NIM, drafts role-specific assessment items from one section at a time. Every draft is routed to a subject-matter expert (SME), who approves or rejects it under an electronic signature that is cryptographically bound to a hash of the approved content. Only approved items are served to learners.

The principal technical contribution is the adaptive engine. Because each generated question retains a foreign-key reference to the SOP section from which it was produced, every learner response constitutes evidence about a specific passage rather than about the document as a whole. The implemented adaptive engine is a hybrid: a rule-based priority layer operating on an exponentially recency-weighted accuracy statistic, supported by two model-based components that estimate parameters online — an Elo rating system for learner ability and item difficulty, and the FSRS memory model for review scheduling. The large language model performs content generation only and takes no part in any adaptive decision.

The current implementation passes 221 backend tests with zero failures, with database migration and deployment checks clean and the frontend building without lint warnings. The complete workflow has been verified end to end against the live model provider. In a verified controlled demonstration, a learner scoring 33.33% on a nine-question assessment received six targeted questions rather than nine, with the section already mastered excluded, and improved from 50% to 100% on both weak sections — a gain of 50 percentage points measured from stored responses. This is a controlled demonstration result and is not presented as a statistical user study.

Keywords: *adaptive learning; spaced repetition; FSRS; Elo rating; large language models; human-in-the-loop; pharmaceutical SOP training; explainable recommendation.*

2. Introduction

Pharmaceutical manufacturing operates under good-practice (GxP) regulations that require personnel to be trained, and periodically retrained, on the Standard Operating Procedures governing their work. The training record is itself a controlled artefact: an organisation must be able to demonstrate who was trained, on what content, and when.

2.1 Limitations of static training

Conventional delivery pairs a fixed set of slides with a single assessment issued identically to every learner. Two consequences follow. First, the assessment yields only an aggregate score; a result of 60% identifies the individual who is struggling but not the material they are struggling with. Second, because

the deficiency is not localised, retraining necessarily repeats the entire procedure, including sections the learner has already demonstrated competence in. This is inefficient for the learner and uninformative for the training coordinator.

2.2 The role of generative AI, and its limits

Large language models can draft assessment items from source text at negligible marginal cost, which makes per-section assessment economically feasible where manual authoring would not be. However, a language model conditioned on a passage produces plausible text; it does not verify that the answer it designates as correct is in fact correct. A recent systematic review of 71 studies reports error rates for AI-generated multiple-choice questions ranging from 0.3% to 45%, and concludes that current evidence does not support unsupervised use in summative assessment [7]. Introducing generative AI into controlled training content therefore introduces a quality risk that must be explicitly controlled.

2.3 Motivation for SME validation and adaptive learning

These two observations motivate the design adopted. Because the model cannot validate its own output, a qualified human reviewer is retained as the sole correctness control, and the architecture is arranged so that no generated content can reach a learner without passing that gate. Because per-section assessment is now feasible, learner responses can be attributed to specific passages, which in turn makes it possible to select future training content on the basis of measured, localised weakness rather than an aggregate score.

3. Problem Statement

The problem addressed by the project comprises the following elements.

- Training content is generic: the same material and the same assessment are issued to every learner irrespective of prior performance.
- Personalisation is absent: an aggregate score does not localise a knowledge gap to a specific part of a procedure.
- Assessment authoring is manual, which limits the number of items and therefore the granularity at which competence can be measured.
- Automated generation introduces reliability concerns, since a language model provides no guarantee of factual correctness.
- Training decisions in a regulated environment must be traceable: it must be possible to state why a particular learner was assigned particular content.
- Content used for assessment must remain controlled, with a qualified individual accountable for its approval.

Formal problem statement

How can SOP-based training be automated while remaining human-validated, explainable, and adaptive to individual learner performance?

4. Aim

To develop an AI-assisted, GxP-oriented training platform that converts pharmaceutical Standard Operating Procedures into SME-reviewed assessments, and that adaptively identifies and retrains learner-specific knowledge gaps at the level of individual procedure sections.

Scope note. The term *GxP-oriented* is used deliberately throughout this report. The platform implements a number of technical controls of the kind associated with 21 CFR Part 11 — identity re-verification at approval, content-bound electronic signatures, immutability of approved records, role-based access control, and an attributed audit trail. It is not a validated system and no claim of regulatory compliance is made. Controls that would be required for such a claim, including tamper-evident audit storage, separation of duties, a qualification record and a formal validation exercise, are not implemented and are identified as such in Section 16.

5. Objectives

The following objectives were defined at the outset. All seven are implemented in the current build; the evidence for each is given in Section 15.

No.	Objective	Description
1	SOP document ingestion	Upload of controlled documents with server-side validation of file type and size.
2	Intelligent chunking	Division of the document into sections using a three-tier cascade, with the tier used recorded per chunk.
3	AI question generation	Section-grounded drafting of role-specific items, with validation, deduplication and a deterministic fallback.
4	SME validation and electronic signature	Human approval with password re-verification, content-hash binding and immutability of approved items.
5	Learner assessment and performance tracking	Server-side grading and maintenance of per-section mastery state.
6	Adaptive retraining	Selection of future content from measured, section-level weakness.
7	Spaced repetition and explainability	Scheduling of review timing by a memory model, with the evidence for every recommendation exposed to the learner.

6. Literature Review

The literature consulted falls into three groups: learner modelling and knowledge tracing, spaced repetition and memory modelling, and the quality of AI-generated assessment content. References were obtained through external academic search and each was verified against the publisher record; no citation in this report is unverified.

Author / Year	Approach	Key contribution	Limitation	Relevance to this project
Corbett & Anderson (1994) [1]	Bayesian Knowledge Tracing	Probabilistic estimation of per-skill mastery from response sequences	Requires many observations per skill; the skill model must be authored manually	Motivates per-section mastery modelling; rejected for the present data scale
Piech et al. (2015) [2]	Deep Knowledge Tracing	Recurrent neural modelling of learner state from interaction sequences	Requires thousands of interactions per learner; the resulting estimate is not interpretable	Explicitly not adopted, on grounds of data scale and auditability
Pelánek (2016) [3]	Elo rating in adaptive education	Joint online estimation of learner skill and item difficulty by treating a response as a match	Coarser than Item Response Theory; no discrimination parameter	Implemented directly (learner K = 32, question K = 16)
Settles & Meeder (2016) [4]	Half-life regression	Trainable spaced-repetition model fitted from large-scale review logs	Depends on a large proprietary review corpus	Informs the difficulty weighting applied to the mastery signal
Ye, Su & Cao (2022) [5]	SSP-MMC scheduling	Formulates review scheduling as a stochastic shortest-path problem	Optimises review cost; does not address content selection	Supports the separation of scheduling from selection
Su, Ye et al. (2023) [6]	DSR memory model (FSRS)	Difficulty–Stability–Retrievability model of memory dynamics	Published weights fitted on approximately 220 million reviews, not per user	Implemented as the review scheduler
Kiyak et al. (2026) [7]	Systematic review of AI-generated MCQ validity	Synthesis of 71 studies across 24 countries	Reported error rates range from 0.3% to 45%; distractor quality frequently weak	Evidence base for a mandatory SME approval gate

Two conclusions were drawn. First, an interpretable online estimator of ability and difficulty is preferable to a trained sequence model at the data volumes a training deployment produces, and is also more defensible in a regulated setting where a recommendation may be questioned. Second, the reported variability in AI-generated assessment quality is sufficiently wide that human review must be a mandatory gate rather than an optional safeguard.

7. Existing System

Conventional SOP training, whether paper-based or delivered through a learning management system, exhibits a consistent structure. A fixed body of material is issued to all personnel in a role. A single assessment, authored manually and revised infrequently, is administered to everyone. A pass mark is recorded against the learner, and periodic retraining is triggered by a fixed calendar interval.

The consequences relevant to this project are as follows.

- Static material: the content does not vary with the learner's demonstrated competence.
- Generic assessment: because every learner receives the same items, the result carries no information about which section of the procedure was misunderstood.
- Manual authoring: item creation is expensive, which constrains assessments to a small number of questions and therefore to coarse measurement.
- Limited personalisation: retraining repeats the whole procedure, since there is no basis on which to omit part of it.

- Weak feedback loop: the assessment result does not influence the content of the next training cycle, and scheduling is by fixed interval rather than by evidence of forgetting.

8. Proposed System

The proposed system replaces the static cycle with a closed loop in which measured performance determines subsequent content. The overall flow is as follows.

SOP → Document processing → Chunking → AI question generation → SME review → Electronic signature

→ Approved question → Learner quiz → Server-side scoring → Section-level mastery

→ Adaptive engine → Targeted retraining → Reassessment → Learning gain → FSRS review scheduling

The proposed system improves on static training in five respects.

1. Content is grounded in the source procedure. Each item is generated from a single section, and the identity of that section is stored with the item, so the provenance of every question is recoverable.
2. Correctness remains under human control. Approval is a mandatory gate; the generative component drafts content and does not decide what is administered.
3. Measurement is localised. Because responses are attributable to sections, competence is measured per section rather than per document.
4. Selection is adaptive. Sections in which the learner has demonstrated competence are excluded from the next assessment, so retraining is proportionate to measured need.
5. Timing is modelled rather than fixed. Review intervals are derived from a memory model fitted to the learner's own responses instead of a uniform calendar rule.

9. System Architecture

The platform is implemented as a Django REST Framework backend with a React single-page frontend. The backend is organised into seven applications separated by domain.

Layer	Technology	Responsibility
Frontend	React 18, Vite	Single-page interface for administrators, SME reviewers and learners; navigation gated by role
Backend	Django 5, Django REST Framework	Seven applications: accounts, sops, quiz, ai_engine, attempts, analytics, audit
Database	PostgreSQL or SQLite, selected by DATABASE_URL	Documents, chunks, questions, attempts, per-section mastery state, audit records

Layer	Technology	Responsibility
Asynchronous processing	Redis and Celery	Three tasks; invoked synchronously, providing process isolation rather than asynchronous execution
AI layer	NVIDIA NIM	meta/llama-3.1-8b-instruct for generation; nvidia/nv-embedqa-e5-v5 for semantic chunking
Document processing	PyMuPDF and internal services	Text extraction and the three-tier chunking cascade
Adaptive engine	attempts/adaptive.py	Determines which sections require retraining
Scheduling	attempts/fsrs.py	Determines when a section should next be reviewed
Authentication and RBAC	DRF token authentication	Administrator, SME reviewer and learner tiers; querysets scoped by role
Audit and security	audit application	Seventeen attributed action types; rate limiting; electronic-signature verification

9.1 Architectural principle: separation of concerns

Two design decisions govern the architecture. The first is that the generative component is confined to content drafting; it does not participate in grading, in mastery estimation, or in any adaptive decision. The second is that the two questions the adaptive system must answer — which material to train, and when to review it — are answered by separate modules, for reasons developed in Section 11.4.

10. Development Methodology and Progress

Development proceeded in ten phases over the period 13 July 2026 to 14 August 2026. The dates given below are taken from the project's version-control history and are not reconstructed. Each phase is identified by the commit that concluded it.

Phase	Date	Commit	Work completed
1 — Requirements and core platform	13 Jul 2026	d8070a6	Problem definition, architecture, authentication, role model, SOP management, database schema, initial SOP-to-quiz pipeline
2 — SME workflow and signatures	21 Jul 2026	7690072	Electronic signatures, model confidence scoring, first adaptive-retraining suggestions
3 — Adaptive retraining (first form)	21 Jul 2026	cdf94b8	Automatic assignment of retraining, initially at whole-SOP granularity
4 — Retrieval chatbot and review preparation	21 Jul 2026	b31f064, da3dbec	Retrieval-augmented SOP chatbot; first review deliverables
5 — Online estimation and scheduling	26 Jul 2026	b5be386	Elo rating for learner ability and item difficulty; FSRS replaced the fixed Leitner interval ladder
6 — Section-level mastery	4 Aug 2026	7b98140	ChunkMastery introduced, moving the adaptive unit from the document to the individual SOP section
7 — Requirements audit	9–14 Aug 2026	d783815, 0e07cf2	Reverse-engineered SRS with per-requirement status; gap analysis; requirements traceability
8 — Adaptive redesign and hardening	14 Aug 2026	0e07cf2	Recency-weighted accuracy, evidence gate, FSRS reconciliation, learning-gain measurement, security hardening, documentation
9 — Final verification and fix	14 Aug 2026	4810223	Independent adversarial audit; duplicate question-identifier defect identified and corrected; frontend lint cleared
10 — Final validation	14 Aug 2026	06b6e95	Full end-to-end demonstration against the live provider; review materials produced

The sequence is worth noting as a methodological point. Phases 1 to 6 delivered functionality. Phase 7 did not add features at all: it produced a Software Requirements Specification by reverse-engineering the implemented system and recording, for each of 84 numbered functional requirements, whether it was implemented and where the supporting code was located. Phases 8 and 9 were then driven by the gaps that audit exposed. The defects described in Section 12 were found in this way rather than reported by users.

11. Adaptive Learning Methodology

This section describes the implemented adaptive engine in detail, as it constitutes the principal technical contribution of the project.

11.1 The unit of adaptation

The adaptive unit is the SOPChunk — a single section of a procedure — and not the document. Each generated question retains a foreign-key reference to the chunk from which it was produced. Consequently a learner response is evidence about a specific passage of a specific procedure. Without

this reference the only computable quantity would be a percentage per document, which is precisely the limitation of existing practice identified in Section 7. The single raw signal consumed by the engine is the boolean correctness of each recorded answer; every other quantity is derived from it.

11.2 Decision pipeline

Learner answers → Performance statistics → Recency weighting → Evidence threshold
 → Mastery state → Priority classification → Question selection → Targeted
 retraining → Reassessment

11.2.1 Recency-weighted performance

For each section, the learner's response history is ordered most-recent-first and an exponentially recency-weighted accuracy is computed, the weight of the i -th most recent response being $0.5^{(i/5)}$. The half-life of five responses matches the window used for the recent-accuracy figure displayed in the interface, so that the term "recent" denotes the same quantity throughout the system.

This replaced an unweighted lifetime average, and the substitution corrected an observable defect. A learner who answered five questions incorrectly and subsequently five correctly has demonstrably improved, yet the lifetime average remains 50%; the interface therefore displayed a recent accuracy of 100% adjacent to a HIGH priority indication and contradicted itself. Under recency weighting the same history yields 66.7% and the section moves to MEDIUM priority. The reverse sequence yields 33.3%, so deterioration is also detected earlier than a lifetime average would permit. Neither transition is instantaneous: a single correct response cannot erase a history of failure, which is the appropriate behaviour for a compliance training record.

11.2.2 Evidence sufficiency

A minimum-evidence threshold of three responses is applied before a section may be excluded from retraining. The rule is deliberately asymmetric: strong performance on fewer than three responses is capped at MEDIUM priority and cannot exclude a section, whereas weak performance on a small sample is still classified as HIGH. The asymmetry reflects the relative cost of the two possible errors — unnecessary retraining costs a small number of additional questions, whereas omitted retraining may leave an operator unqualified.

11.2.3 Priority classification

Classification proceeds in a fixed order, and the order is itself significant.

6. Never assessed. A section with no recorded responses is classified HIGH and evaluated first, so that absence of evidence is never interpreted as competence.
7. Mastered. A section with three consecutive passing assessments is assigned priority NONE and retired from selection.
8. Insufficient evidence. Strong performance below the minimum-evidence threshold is capped at MEDIUM.
9. Threshold comparison. Below 60% is HIGH; below 80% is MEDIUM; at or above 80% is LOW.

Parameter	Value	Role
REGENCY_HALF_LIFE	5 responses	Exponential decay applied to older responses
MIN_EVIDENCE	3 responses	Minimum evidence required to exclude a section
WEAK_ACCURACY_THRESHOLD	60%	Boundary below which a section is HIGH priority
PROFICIENT_ACCURACY_THRESHOLD	80%	Boundary below which a section is MEDIUM priority
MASTERY_STREAK_THRESHOLD	3 assessments	Consecutive passes required for mastery

These values are documented chosen defaults rather than fitted parameters. The 80% threshold matches the pass mark already used elsewhere in the system. With operational data, MIN_EVIDENCE would be the first candidate for tuning. *Classification is applied to the accuracy figure as displayed, rounded to one decimal place, so that the recorded verdict cannot contradict the percentage shown alongside it; this is a deliberate trade-off and is documented in the source.*

11.3 Online estimation components

11.3.1 Elo — learner ability and item difficulty

Following Pelánek [3], each response is treated as a match between the learner's current ability rating and the item's current difficulty rating, both initialised at 1500. Two update rates are used: $K = 32$ for the learner, because an ability estimate must adapt from a small number of responses, and $K = 16$ for the item, because an item's rating is shared across all learners and should not be displaced by a single individual. Elo output weights the pass signal that drives mastery and informs the suggested difficulty of subsequent assignments. At section level an ability-only update is applied, since the whole-attempt update has already adjusted item difficulty once and repeating it would count a single response twice.

11.3.2 FSRS — memory state and review timing

Review scheduling uses the Difficulty–Stability–Retrievability memory model [6], with the seventeen published weights and a target retention of 0.9. The system records stability and difficulty per section and schedules the next review at the point where predicted recall falls to the target. Only two of the four FSRS grades are used, since the available signal is binary; using the model honestly on the signal available was preferred to inventing a four-point scale. The earlier fixed Leitner interval ladder was replaced by this model in Phase 5; the Leitner box index is still computed and displayed as an audit signal but no longer determines the schedule.

11.4 Separation of "what" from "when"

The adaptive engine and the scheduler answer different questions and are implemented as separate modules. The adaptive engine determines what should be trained; FSRS determines when it should be reviewed. The separation is not stylistic but necessary.

FSRS retrievability $R(\text{elapsed}, \text{stability})$ evaluates to 1.0 for every value of stability when elapsed time is zero. Immediately after an assessment — precisely the moment at which the retraining decision must be made — a section the learner has just failed and a section the learner has just passed both return 1.0. Retrievability therefore cannot discriminate between them, and cannot be used to select content.

Observed accuracy does discriminate between the two cases and is used for selection; FSRS retains responsibility for timing. The interface reports both quantities, so that a section may be shown as requiring review while simultaneously being scheduled for a future date. An earlier implementation reported only the adaptive verdict, with the consequence that a section could be recommended while the assessment screen had nothing to offer; this is recorded as Defect 8 in Section 12.

11.5 Classification of the implemented system

For accuracy, the implemented adaptive engine is characterised as a hybrid system of four layers.

Layer	Mechanism	Parameters estimated from learner data?
1 — Adaptive priority	Rule-based: fixed thresholds, evidence gate, mastery state	No — values are chosen, not fitted
2 — Performance signal	Statistical: exponentially recency-weighted accuracy	Yes — the statistic is computed from responses
3 — Elo	Model-based online parameter estimation	Yes — learner ability and item difficulty update per response
4 — FSRS	Model-based memory model	Yes for per-section state; the seventeen weights are fixed published defaults

Statements of scope

The large language model does not determine adaptive priority. Its role is confined to drafting candidate assessment items.

The adaptive engine is not a trained neural network. No model is trained offline; there is no training dataset, no loss function and no gradient-based optimisation.

The system is adaptive in the technical sense that measured learner performance changes the content selected for subsequent assessment, and two of its components update their parameters online from learner responses.

12. Defects Identified and Corrected During Development

The defects recorded below were identified through scenario-based testing and a structured audit of the implementation against its own specification, rather than through user reports. Each is presented with its root cause, its impact, the correction applied and the validation that now protects against regression. The first three concern the correctness of the adaptive engine; the next three concern security and data integrity; the last two were identified during final verification.

Defect 1: Correctly answered sections were still selected for retraining

Aspect	Description
Problem	Retraining selection consumed only the binary mastery status, which changes state only after a full streak of passing assessments has accumulated. A section answered correctly was therefore indistinguishable from one answered incorrectly until that streak completed.
Root cause	Selection was driven by a streak counter rather than by observed response accuracy.
Impact	Early retraining cycles replayed the entire procedure, which is the opposite of the intended adaptive behaviour and reproduces the deficiency of static training.
Solution	Selection was reimplemented to rank sections by observed accuracy, so that a demonstrated section is excluded immediately rather than after a delay.
Validation	Scenario tests assert that a section answered correctly is excluded from the subsequent selection.

Defect 2: Sections never assessed were invisible to selection

Aspect	Description
Problem	A section that had never been assessed had no mastery record, and selection iterated over existing mastery records only. Such sections could therefore never be selected.
Root cause	Absence of data was represented by absence of a record, and the absent record was interpreted as nothing to do.
Impact	A learner assessed only on one part of a procedure would be retrained only on that part indefinitely, and the remainder of the procedure would never be assessed.
Solution	Never-assessed sections are now treated as an explicit state and classified HIGH priority, evaluated before any other check.
Validation	A regression test asserts that a section with no recorded responses is returned at HIGH priority.

Defect 3: Whole-document mastery concealed weak sections

Aspect	Description
Problem	Mastery at the level of the whole SOP gated selection. Three passing assessments covering one section marked the entire document mastered.
Root cause	A single coarse-grained state was used to gate a decision that required finer granularity.
Impact	A document could be recorded as mastered while a section within it was still being failed, and retraining returned no assignments at all.
Solution	Section-level mastery (ChunkMastery) was introduced in Phase 6, and document-level mastery now defers to measured section-level evidence.
Validation	A scenario test constructs precisely this situation and asserts that the failing section is still selected.

Defect 4: The answer key was exposed to learners

Aspect	Description
Problem	The question serialiser returned the correct-option flag and the explanation to every authenticated caller, including learners taking an assessment.
Root cause	A single serialiser served both the reviewer interface, which legitimately requires the answer key, and the learner interface, which must not receive it.
Impact	Assessment results were not trustworthy, and consequently neither was any mastery state or adaptive decision derived from them.
Solution	Separate serialisers were introduced and are selected by role. The learner serialiser omits the correct-option flag and the explanation entirely, and the related options endpoint was restricted to reviewers.
Validation	Eleven tests, including an assertion against the raw response body rather than the parsed object.

Defect 5: A completed assessment attempt could be resubmitted

Aspect	Description
Problem	No guard prevented a second submission against an attempt that had already been completed and graded.
Root cause	Completion was recorded but never checked before grading, and a read-then-check sequence would in any case have been vulnerable to concurrent submissions.
Impact	A completed training record could be overwritten, which is unacceptable for a record intended to evidence competence.
Solution	The attempt is claimed by an atomic conditional update, so that only one submission can succeed. The losing submission receives HTTP 409 and the event is recorded in the audit trail. The approach does not rely on row-level locking and is therefore correct on all supported database backends.
Validation	Tests assert a 409 response on resubmission and that the stored score is unchanged.

Defect 6: Approved content could be modified after signature

Aspect	Description
Problem	The restriction preventing modification of an approved question was enforced only in the client interface; the API accepted the write.
Root cause	A control was implemented in presentation rather than at the point of enforcement.
Impact	Content could be altered after approval, which would silently invalidate the electronic signature bound to it.
Solution	Update, partial-update and delete operations on an approved question now return HTTP 403. Approval stores a SHA-256 hash over the canonical representation of the approved content, and a verification routine recomputes that hash on demand to detect modification performed by any other route.
Validation	Fourteen tests covering the signature path, including detection of out-of-band modification.

Defect 7: Duplicate question identifiers fabricated evidence

Aspect	Description
Problem	A submission containing the same question identifier more than once was accepted, and each repetition was stored as a separate response record.

Aspect	Description
Root cause	Submission validation checked that each identifier was permissible but not that the set of identifiers was distinct.
Impact	Submitting a single known question several times wrote several correct responses into that section's history. This was sufficient to satisfy the minimum-evidence threshold with one item and to raise both the recency-weighted accuracy on which the adaptive engine decides and the section's ability rating. Measured before correction: the same identifier submitted five times produced five response records and moved the item rating from 1500 to 1570.
Solution	Submissions containing duplicate identifiers are rejected with HTTP 400 before any database write, alongside the existing payload checks and before the attempt is claimed, so that a rejected submission leaves the attempt open and retakeable.
Validation	Two tests, covering three, five and ten repetitions, asserting rejection with no response records, no mastery record, no rating movement and the attempt not consumed. Verified independently on an isolated database.

Defect 8: Recommendations could not be acted upon

Aspect	Description
Problem	The learning path reported sections requiring retraining without regard to whether the scheduler considered them due, so a section could be recommended while the assessment screen had nothing to offer.
Root cause	The adaptive verdict and the scheduling constraint were computed independently and only the former was surfaced.
Impact	The interface promised training that the assignment mechanism would not deliver.
Solution	Each section now reports both its selection status and its due status, and the two are combined into an availability flag. The interface distinguishes sections requiring review from those available immediately, labelling the remainder with their scheduled date.
Validation	Integration tests assert consistency between what the learning path reports and what the assignment mechanism returns.

Three further improvements were made in response to the same audit and are recorded here for completeness: the substitution of recency-weighted accuracy for a lifetime average (Section 11.2.1); the introduction of the minimum-evidence threshold (Section 11.2.2); and the addition of measured learning gain, which compares the older half of a section's response history against the more recent half.

13. Testing and Validation

The verification results reported in this section were obtained by executing the project's test suite and verification commands in the current build. The backend suite comprises 221 tests, all passing, with no failures and none skipped.

13.1 Test distribution by application

Application	Tests	Principal coverage
attempts	99	Adaptive classification, mastery, Elo, FSRS, submission validation, learning path
quiz	36	Role-based serialisation, approval workflow, electronic signature, immutability
ai_engine	36	Generation, structured-output validation, deduplication, fallback, error classification
sops	19	Upload validation, extraction, chunking cascade, audited deletion
accounts	16	Authentication, role resolution, rate limiting
analytics	10	Dashboard scoping and aggregation
audit	5	Audit record creation and retrieval
Total	221	All passing; 0 failures; 0 skipped

13.2 Categories of testing performed

- Unit testing of the adaptive policy functions, the Elo update rules and the FSRS memory-model computations in isolation.
- Integration testing of the complete submission path, covering grading, mastery update, rating update and scheduling within a single transaction.
- Security regression testing covering answer-key exposure, resubmission, cross-document submission, unapproved content, duplicate identifiers, modification of approved content and access scoping.
- Adaptive scenario testing, described in Section 13.3.
- API validation covering rejection paths and the associated status codes.
- Frontend verification: ESLint reports zero errors and zero warnings, and the production build completes successfully.
- End-to-end verification of the complete workflow against the live model provider.
- Static verification: the database migration check reports no outstanding changes and the deployment configuration check reports no issues.

13.3 Controlled adaptive scenarios

Fourteen controlled scenarios were executed against the implemented policy functions to confirm that classification behaves as specified across the range of learner states. All fourteen produced the expected classification.

Scenario	Condition	Expected priority	Result
Never assessed	No recorded responses	HIGH	As expected
Strong learner	Five of five correct	LOW	As expected
Weak learner	Zero of five correct	HIGH	As expected
Insufficient evidence (strong)	One or two correct responses	MEDIUM (capped)	As expected
Insufficient evidence (weak)	One or two incorrect responses	HIGH	As expected
Evidence threshold met	Three correct responses	LOW	As expected
Improving learner	Five incorrect then five correct	MEDIUM (66.7%)	As expected
Continued improvement	Five further correct responses	LOW (85.7%)	As expected
Declining learner	Five correct then five incorrect	HIGH (33.3%)	As expected
Mastered	Three consecutive passing assessments	NONE	As expected
Mastery lost	Failure following mastery	HIGH	As expected
Mixed performance	Alternating correct and incorrect	MEDIUM	As expected
Multiple learners	Two learners, differing profiles, same document	Different question sets	As expected
Duplicate submission	Same identifier submitted three, five and ten times	Rejected, no writes	As expected

In addition, an exhaustive check was performed over every possible response sequence of length three to fourteen, confirming that the classification is monotonic: an additional correct response never raises a section's priority, and an additional incorrect response never lowers it.

14. Results

The results in this section are obtained from a verified controlled demonstration executed against the live model provider. They demonstrate that the implemented workflow operates end to end and that the adaptive mechanism behaves as designed. They are not a statistical user study, and no claim of population-level effectiveness, statistical significance or clinical validation is made or implied.

14.1 Workflow verification

Each stage of the workflow was exercised and confirmed in a single continuous execution.

Stage	Outcome
SOP ingestion and processing	Document processed; three sections produced by heading-aware chunking

Stage	Outcome
AI question generation	Nine questions generated by the live provider (meta/llama-3.1-8b-instruct), three per section; no duplicates skipped
SME review and electronic signature	Nine questions approved; signature bound to content hash; integrity verification returned intact
Learner assessment	First attempt scored 33.33% (three of nine correct)
Server-side scoring and mastery update	Per-section mastery updated; ability ratings separated the strong section from the two weak sections
Adaptive analysis	Two sections classified HIGH with stated evidence; one section classified LOW
Learning path explainability	Each recommendation reported the adaptive score, lifetime accuracy, response count and the threshold applied
Targeted retraining	Six of nine questions selected; the section already demonstrated was excluded
Reassessment	Retraining assessments completed at 100%
Learning gain	Both weak sections improved from 50% to 100%
FSRS scheduling	Next review dates computed per section; weak sections scheduled earlier than the strong section

14.2 Adaptive selection

Measure	Value
Questions generated	9 (three sections, three per section)
First-attempt score	33.33% (three of nine)
Questions selected for retraining	6 of 9
Sections excluded	1 (competence already demonstrated)
Sections targeted	2 (both classified HIGH)

14.3 Measured learning gain

Learning gain is computed per section by comparing the older half of the learner's recorded response history for that section against the more recent half, and is reported only where at least four responses exist. The values below are computed from stored responses; they are measured rather than asserted.

Section	Before retraining	After retraining	Gain
Documentation Practices	50%	100%	+50 percentage points
CAPA and Root Cause Analysis	50%	100%	+50 percentage points
Good Manufacturing Practice	100%	100% (excluded from retraining)	Not applicable

A secondary observation supports the design decision described in Section 11.2.1: following retraining the recency-weighted score for both targeted sections was 87.9% while the unweighted lifetime

accuracy was 75.0%. The divergence is the recency weighting registering improvement that a lifetime average would continue to suppress.

14.4 System verification results

Check	Result
Backend test suite	221 tests, 0 failures, 0 skipped
Database migration check	No outstanding changes detected
Deployment configuration check	No issues identified
Frontend static analysis (ESLint)	0 errors, 0 warnings
Frontend production build	Completed successfully
End-to-end workflow	Verified against the live model provider

15. Current Project Status

The table below records the status of each principal capability together with the evidence supporting that status.

Feature	Status	Validation
SOP ingestion and validation	Completed	Verified; unit tested
Text extraction and three-tier chunking	Completed	Verified; strategy recorded per chunk
AI question generation	Completed	Verified against live NVIDIA NIM
Structured-output validation and deduplication	Completed	Unit tested (lexical deduplication)
Deterministic offline fallback	Completed	Unit tested
Source-chunk provenance	Completed	Regression tested; exposed in the reviewer interface
SME review workflow	Completed	Tested
Electronic signature and content hash	Completed	Tested, including tamper detection
Immutability of approved content	Completed	Tested (HTTP 403 on write)
Learner assessment and server-side scoring	Completed	Tested
Answer-key protection	Completed	Tested, including raw-response assertion
Submission validation	Completed	Tested (document, role, approval, duplicates)
Single-submission guarantee	Completed	Tested (HTTP 409, atomic claim)
Section-level mastery	Completed	Scenario tested
Adaptive priority classification	Completed	Fourteen controlled scenarios
Recency-weighted performance	Completed	Unit and integration tested
Elo ability and difficulty estimation	Completed	Unit and integration tested
FSRS review scheduling	Completed	Integration tested
Learning-gain measurement	Completed	Verified in demonstration
Explainable learning path	Completed	Interface and API verified
Role-based access control	Completed	Tested
Audit trail	Completed	Seventeen action types; tested
Rate limiting	Completed	Tested
Retrieval-augmented SOP chatbot	Completed	Implemented in Phase 4
Server-side offered-question binding	Not implemented	Future scope (Section 16.1)
SOP version lifecycle	Not implemented	Future scope; reprocessing is blocked, not versioned
Question revision history	Not implemented	Future scope
Difficulty-aware adaptive priority	Not implemented	Future scope
Entailment verification of generated content	Not implemented	Future scope
Tamper-evident audit storage	Not implemented	Future scope
Separation of duties	Not implemented	Future scope

Feature	Status	Validation
Frontend automated test suite	Not implemented	Verified by lint, build and manual checks only

15.1 Objectively derived completion measure

A completion figure is reported here only because the project's Software Requirements Specification enumerates 84 numbered functional requirements, each carrying a recorded implementation status and a reference to the supporting code. The figure below is derived by counting those statuses and is not an estimate.

Measure	Count	Proportion
Functional requirements specified	84	100%
Implemented at the Phase 7 audit baseline	76	90.5%
Closed during Phases 8–9 (FR-085, FR-096, FR-097, FR-145)	4	+4.8%
Implemented in the current build	80	95.2%
Remaining unimplemented (FR-045, FR-060, FR-146, FR-152)	4	4.8%

The four requirements closed after the audit are: server-side locking of approved questions (FR-085), prevention of resubmission of a completed attempt (FR-096), validation that submitted questions belong to the attempt and are approved (FR-097), and auditing of deletions and content edits (FR-145). The four that remain open are the capture of page numbers on chunks (FR-045), caller-specified generation difficulty (FR-060), tamper-evident audit storage (FR-146), and administrative registration of the mastery models (FR-152).

This measure counts numbered functional requirements only. It should not be read as a measure of overall project maturity: the unimplemented items described in Section 16 include capabilities, such as document version control, whose absence is more significant than their single-requirement weight suggests.

16. Limitations

The limitations below are properties of the current build. They are stated as identified opportunities for subsequent development rather than as defects, and each has an identified remedy.

16.1 The offered question set is not persisted

The set of questions selected by the adaptive engine for a given attempt is computed server-side but is not recorded. Validation confirms that every submitted question belongs to the attempt's document, matches the learner's role, is approved and is not duplicated; it cannot confirm that the question was among those offered. A modified client could therefore submit an approved question that the engine had excluded, or omit questions — and because the score is computed over submitted responses, omission inflates it. The exposure is bounded to the learner's own attempts, within their permitted

document and approved content, and every submission is audited; the conforming client always submits the complete offered set. The adaptive decision is accordingly computed and validated server-side but is not yet enforced. A second consequence is that a past attempt cannot be reconstructed exactly as it was administered, which weakens the training record. The remedy is a persisted association between attempt and offered question, written at attempt creation and checked at submission.

16.2 No SOP version lifecycle

Reprocessing a document that has approved questions is refused with HTTP 409. This prevents data loss — reprocessing would delete the document's chunks, remove every learner's section-level mastery by cascade, and detach approved questions from their source text — but it constitutes a guard rather than a workflow. There is therefore no supported path for revising a procedure, which in a regulated context is an operational limitation. The remedy is a version entity interposed between document and chunk, so that publication of a revision leaves the prior version's chunks, questions and mastery intact.

16.3 Difficulty does not influence adaptive priority

Item difficulty weights the pass signal that drives mastery and scheduling, but does not enter the accuracy computation used for priority classification. An incorrect response to an easy item and an incorrect response to a difficult item therefore contribute equally to the determination that a section is weak. The remedy is to weight each response by its item's current rating within the recency-weighted accuracy calculation.

16.4 Grounding is provenance, not entailment

Generated content is constrained by supplying a single section per prompt and by instructing the model not to draw on outside knowledge, and the identity of the source section is stored with each item and displayed to the reviewer. No mechanical verification establishes that the designated correct answer is entailed by that passage. Grounding is therefore provenance together with a prompt constraint, and not verified entailment; the SME review remains the only correctness control. The remedy is an entailment or judge model scoring support of the answer by its source passage, surfaced to the reviewer as an additional signal rather than as a replacement for review.

16.5 No formal knowledge-tracing model

Mastery is estimated by a rule-based classification over a recency-weighted statistic, supported by Elo and FSRS. No Bayesian Knowledge Tracing or Item Response Theory model is implemented. This reflects the data volume a training deployment produces: Item Response Theory in particular requires substantially more responses per item than the present deployment generates before item parameters can be calibrated reliably.

16.6 Further limitations

- Deduplication of generated items is lexical and does not detect semantically equivalent items expressed in different vocabulary.
- The audit trail is append-only by convention and administrative configuration; it is not tamper-evident, as no hash chaining or write-once storage is implemented.

- Separation of duties is not enforced: an administrator may both generate and approve content.
- Editing a rejected item and re-approving it overwrites the superseded wording, so full content lineage is not retained.
- The frontend has no automated test suite and is verified by static analysis, successful build and manual checking.
- Asynchronous tasks are dispatched and awaited synchronously, providing process isolation but not asynchronous execution.
- The system has not been deployed; container and continuous-integration configurations exist but have not been executed as complete stacks.

17. Future Scope

Planned work is divided into three tiers. The ordering within the short-term tier is deliberate and reflects a principle of integrity first, then quality of measurement, then additional capability.

17.1 Short term

Priority	Item	Rationale
1	Persisted attempt–question association	Converts the adaptive decision from advisory to enforced and makes past attempts reproducible
2	SOP version lifecycle	The only remaining gap capable of silently destroying learner mastery data
3	Question revision history	Completes content lineage; the electronic signature would bind to a version rather than a mutable record
4	Difficulty-aware adaptive priority	Removes the internal inconsistency described in Section 16.3
5	Semantic duplicate detection	Detects equivalent items that share no vocabulary with one another
6	Entailment verification	Upgrades grounding from provenance to verification, informing rather than replacing SME review

17.2 Advanced

- Bayesian Knowledge Tracing, as an interpretable and comparatively sample-efficient probabilistic mastery estimator.
- Item Response Theory, to obtain calibrated item difficulty and discrimination parameters once sufficient responses per item exist.
- Computerised adaptive testing, selecting each item from the preceding response rather than fixing the set in advance; this depends on calibrated item parameters and is therefore gated on the preceding item.
- Prerequisite and concept graphs, so that retraining sequences foundational material before dependent material.
- Cross-document concept modelling, so that competence demonstrated on a concept in one procedure informs assessment of the same concept in another.

17.3 Research and agentic direction

None of the capabilities described in this subsection is implemented in the current build. They are stated as future research directions.

- An adaptive tutor agent capable of conducting a remedial dialogue grounded in the relevant SOP section.
- Agentic learning-path planning, in which a supervised pipeline follows an observe, analyse, plan, retrieve, train, evaluate and adapt cycle with explicit hand-offs and retries between stages.
- Personalised training orchestration across a procedure library, sequencing work across multiple documents for an individual learner.

Constraint on agentic development. Any agentic capability introduced subsequently would operate strictly above the deterministic control layer. It would not replace SME approval, would not modify approved content, and would not participate in grading or in the electronic-signature path. The property that the generative component drafts content while a qualified human decides what is administered is a design invariant of the platform and is not intended to be relaxed.

18. Conclusion

The project has produced a working platform that converts controlled pharmaceutical SOPs into SME-approved assessments and uses learner performance, measured at the level of individual procedure sections, to determine the content of subsequent training. Development proceeded over ten phases between 13 July and 14 August 2026, with the later phases directed by a structured audit of the implementation against its own specification rather than by the addition of further features.

Generative AI is employed for scalable question drafting, constrained to a single source section per prompt and supported by a deterministic offline fallback. Correctness remains under human control through a mandatory review gate in which approval is bound to a cryptographic hash of the approved content and after which that content is immutable. The adaptive engine is a hybrid of a rule-based priority layer, a recency-weighted statistical signal, and two model-based components that estimate parameters online; it is explainable by construction, since every recommendation carries the measured evidence and the threshold that produced it. The question of what to train is deliberately separated from the question of when to review, for the mathematical reason set out in Section 11.4.

The current build passes 221 backend tests with no failures, with migration and deployment checks clean and the frontend building without warnings. Eighty of the eighty-four specified functional requirements are implemented. In a verified controlled demonstration the platform excluded a section the learner had already mastered, retrained only the two weak sections, and recorded an improvement from 50% to 100% on both — a gain measured from stored responses rather than asserted.

The contribution of this project is not the application of a language model to question generation, which is by now routine. It is the demonstration that a probabilistic generative component can be placed inside a controlled training workflow without displacing human accountability, and that the resulting learner data can drive an adaptive decision that remains explainable to a reviewer. Generative AI produces the content; qualified reviewers control it; learner response data drives adaptation; and a memory model governs review timing.

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Appendix A. Progress Summary

Area	Initial state	Current state	Status
Requirements	Informal problem statement	84 numbered functional requirements with recorded status and code references	Completed
Core platform	Planned	Authentication, role model, document management, dashboard	Completed
Document processing	Planned	Extraction with three-tier chunking cascade; strategy recorded per chunk	Completed
Question generation	Planned	Live generation via NVIDIA NIM with validation, deduplication and deterministic fallback	Completed
SME workflow	Not present	Review, approval, electronic signature bound to content hash, immutability	Completed
Assessment	Basic quiz	Server-side scoring, answer-key protection, submission validation, single-submission guarantee	Completed
Adaptive learning	Conceptual	Hybrid engine: rule-based priority over a recency-weighted statistic, with Elo and FSRS	Completed
Adaptive granularity	Whole document	Individual SOP section	Completed
Scheduling	Fixed Leitner interval ladder	FSRS memory model with per-section state	Completed
Explainability	Score only	Per-section evidence, applied threshold and stated reason	Completed
Learning measurement	Not present	Measured gain from stored responses (older half against newer half)	Completed
Security	Baseline authentication	Role-scoped serialisation, submission validation, immutability, audit trail, rate limiting	Completed
Testing	Initial tests (89)	221 tests, 0 failures, 0 skipped	Completed
Document versioning	Not present	Reprocessing blocked with HTTP 409; no version entity	Not implemented
Offered-question binding	Not present	Computed and validated but not persisted	Not implemented

Appendix B. Progress Timeline

Derived from the project's version-control history.

13 Jul 2026 Requirements, architecture and core platform



21 Jul 2026 SME workflow, electronic signatures, first adaptive retraining



21 Jul 2026 Retrieval-augmented SOP chatbot



26 Jul 2026 Elo online estimation; FSRS replaces fixed interval scheduling



4 Aug 2026 Section-level mastery (adaptive unit moves from document to section)



9–14 Aug 2026 Requirements audit and gap analysis (no features added)

↓

14 Aug 2026 Adaptive redesign: recency weighting, evidence gate, FSRS reconciliation

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14 Aug 2026 Security hardening and documentation

↓

14 Aug 2026 Final adversarial verification; duplicate-identifier defect corrected

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14 Aug 2026 End-to-end validation against the live provider